Jimmy Lee

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ACADEMIC POSITIONS

Postdoctoral Fellow in Economics Sept 2024 – Aug 2026
Global Poverty Research Lab, Northwestern University

Visiting Scholar Jan 2025 – June 2026
Center for International Development, Harvard University

OTHER EMPLOYMENT

AI Trainer – Economics Expert (Flex time), xAI Apr 2026 – Present

RESEARCH FIELDS

Development Economics, Program Evaluation and Scaling
Focus—Synergies between Agriculture, Rural Education, and Digital Technologies

Behavioral Economics with applications in program evaluation, agriculture, and education
Focus—Modeling how agents jointly choose how much weight to place on mental model(s) alternative to a default—possibly several at once—and how much effort to invest in acquiring information about a new situation, under neurological constraints

With implications for designing interventions that overcome preconceptions and stimulate curiosity, and for measuring beliefs under (possibly) multiple priors

Family Economics, Organizational Economics
Focus—Intergenerational Dynamics in Rural Household and Community Decision Making

EDUCATION

Ph.D., Economics, Northwestern University 2024
M.Phil., Economics, The Chinese University of Hong Kong 2017
B.SSc., Economics, The Chinese University of Hong Kong 2015

JOB MARKET PAPER

"Activating School-Based Agricultural Extension with Parental Engagement: Experimental Evidence from Liberia"
(Funded by AFD Fund for Innovation in Development, USAID-DIV, Wellspring Foundation, and FAO)

Many African countries have vastly under-staffed agricultural extension systems, while rural households face stark tradeoffs between children's schooling and farm labor. This paper studies a school-based agricultural extension (SBAE) program in Liberia that integrates (i) experiential science instruction and student-centered extra-curricular activities; (ii) school farms as demonstration plots; and (iii) student home farming projects to tackle the dual challenges of diffusing agricultural technologies and motivating school attendance.

I randomly assign SBAE across 197 junior high schools to estimate its intent-to-treat effects on voluntary participants. To tackle weak parent-school linkages, in a random subset of SBAE schools, I engage parents through a promotional video highlighting the complementarities between agriculture and education under SBAE, an annual farmer field day, or both.

By the third year, SBAE successfully trains teachers and students but fails to improve technology adoption on any household parcels—and increases student absenteeism by 31%. Conversely, SBAE augmented with both types of engagement increases adopted techniques by 26% on parents' parcels and 53% on students' parcels. It also reduces student dropouts by 4-5 percentage points and absenteeism by 18%-28% over consecutive years, and increases the proportion of students saving for university by 6 percentage points. Parental engagement is pivotal because it activates muted mechanisms—(i) parents increase (otherwise reduced) school visits, learn techniques, and observe harvests; (ii) students manage household farms, use techniques, show parents improved yields, and increase (otherwise reduced) studying hours. Consistent with parents' confusion of SBAE with school gardens, the video clarifying SBAE's student-centered training changes parents' beliefs and plays a dominant role in diffusing technologies through mechanism (ii) and reducing dropouts. The augmented program is more cost-effective than most extension programs in Africa and over twice as cost-effective as conditional cash transfers in raising attendance.

The findings underscore the promise of positioning schools as rural development hubs that break the cycle of stagnant agricultural productivity and low levels of human capital, and exemplify how strengthening parent-school linkages reveals school programs' hidden potential.

OTHER PAPERS

"Mediating Parents' Beliefs as a Nudge: Empowering Students to Benefit from School-Based Agricultural Extension in Liberia"

(Funded by NSF, Weiss Fund, Global Poverty Research Lab, and the Buffett Institute for Global Affairs)

Developing countries face the dual challenges of diffusing agricultural technologies and motivating school attendance. In a randomized evaluation across 197 junior high schools in Liberia, Lee (2025) shows that schools can be developed into hubs that cost-effectively (i) diffuse technologies to rural households; (ii) improve students' livelihoods; and (iii) increase school attendance—but only when parents are engaged with program information. This paper studies a sub-treatment crucial for (ii) and (iii): exploiting the fact that engaged parents express optimism about students' growth in farm management abilities, the intervention reveals this optimism to empower students to manage household farms.

This signal about parents' openness to delegating farm management is a highly effective nudge. Two years later, SBAE improves students' livelihoods and education only with this nudge: annual income rises by 63%, timely payment of school fees rises by 43-47%, and impact per dollar on attendance doubles. Consistent with students' aspirations, students request and shift time from subsistence parcels to managing sales-oriented parcels, and adopt practices enabling timely income from school-tested crops. Both boys and girls, across grades, benefit—often through different channels. Farm interactions unique to the targeted parent-student dyad pinpoint a key mechanism in intra-household dynamics. Even when parents are engaged and students could infer parents' types from school visits and technology adoption, third-party communication remains pivotal in intergenerational resource allocation.

Paired with effective programs, signaling influential household members' openness to changing resource control offers a low-cost, non-intrusive approach to changing intra-household dynamics.

WORK IN PROGRESS

Behavioral economic theory with applications in program evaluation, agriculture, and education

1. **"Default Mental Model, Alternative(s) Activation, and Information Acquisition under Neurological Constraints: Theory and Belief Measurements"**

Policy paper with a model explaining SBAE's promise in an environment characterized by substantial heterogeneity in suitable agricultural technologies

2. **"Catalyzing Technology Diffusion in Africa through School-Based Agricultural Extension"** (joint with Chris Udry)

Scheduled RCTs on the scaling of SBAE in 2026-29

3. **"Learning to Learn: Synergizing Inquiry-based Science and Agricultural Transformation"** (joint with Vesall Nourani, Harvard GSE)
4. **"Parental Engagement in School-based Agricultural Extension: Testing Mechanisms at Scale"** (piloted interventions in 2025-26)

Long-run scaling of SBAE across Liberia, Côte d'Ivoire, and Kenya in 2026-35

5. **"Directed Technical Change under Customized Farming Advice: An Empirical Framework"** *(grant from Weiss Fund; piloted survey method; planning RCT)*
6. **"Parental Engagement and the Use of Unconditional Cash Transfers"**

INVITED PRESENTATIONS

During postdoctoral studies:

National Chung Cheng University; NYU Africa House–Center for Technology, Economics, and Development (CTED); Mohammed VI Polytechnic University (UM6P); Tel Aviv University;

Oklahoma State University; Harvard Graduate School of Education; Harvard Center for International Development; Africa Food Systems Forum 2024

During PhD:

Meeting of the Society of Economics of the Household (SEHO); J-PAL—Middle East and North Africa (MENA); CGIAR—Standing Panel on Impact Assessment (SPIA); International Food Policy Research Institute (IFPRI); University of Glasgow; National Graduate Institute for Policy Studies (GRIPS, Tokyo); Development Day 2023

GRANTS AND AWARDS

Total program evaluation and academic research grants secured for SBAE in Liberia: **$4.2 million**

Program Evaluation Grants

AFD Fund for Innovation in Development ($1.2 mil; co-PI)	2025
J-PAL Learning for All Initiative: Scaling Award ($222K; co-PI)	2024
USAID Development Innovation Ventures ($395K; co-PI)	2022
Wellspring Foundation ($192K; co-PI)	2022
AFD Fund for Innovation in Development ($1.2 mil; co-PI)	2021
Food and Agriculture Organization ($68K; co-PI)	2021

Academic Research Awards

J-PAL Learning for All Initiative ($400K; co-PI)	2025
Weiss Fund ($5K; Lead PI)	2022
National Science Foundation ($451K; co-PI)	2020
Weiss Fund ($29K; Lead PI)	2020
Global Poverty Research Lab ($19K; Lead PI)	2019
Buffett Institute for Global Affairs ($5K; Lead PI)	2019

TEACHING

Teaching Assistant, Northwestern University

Introduction to Microeconomics, Introduction to Macroeconomics, Intermediate Microeconomics II, Intermediate Macroeconomics, International Trade, Economics of Nonprofit Organizations

LANGUAGES AND SKILLS

Citizenship: Hong Kong Special Administrative Region, China
Languages: English (fluent), French (intermediate), Mandarin (fluent), Cantonese (native)
Programming: Stata, R, Matlab, Python, ArcGIS, SurveyCTO

REFERENCES

Professor Christopher Udry (Chair)
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